

Circuit and working

The circuit diagram of Arduino based liquid vending machine is shown in Fig. 2. This project consists of analogue flow sensor, RFID tag, RFID reader, 16x2 LCD, solenoid, Arduino Uno, a single channel relay and a 12V DC power supply.

Analogue water flow sensor. This sensor measures the flow rate of a fluid. It has two openings—one opening for fluid in-take and the other for out-take.

It works on the principle of the Hall effect. The Hall effect is utilised

in the flow meter using a small fan/propeller-shaped rotor, which is placed in the path of the liquid flow.

It has three wires: red wire for supply voltage, black wire for ground and a yellow wire to collect output from Hall effect sensor. Supply voltage could be from 5V to 18V DC.

RFID reader. EM-18 RFID module (refer Fig. 3) is used to read the RFID tag. It decodes and transmits the signal to Arduino via a serial communication protocol.

LCD. A 16x2 LCD (liquid crystal display) displays the data received from Arduino board.

Solenoid. A 12V solenoid is used to control the flow of liquid. When the solenoid gets energised, it opens its valve and allows fluid to flow through the flow sensor; otherwise it stays closed and does not allow fluid to flow across it. The solenoid valve used in this project is shown in Fig. 4.

Main Components

S.No.	Components
1	RFID tag
2	RFID reader (EM-18)
3	12V solenoid
4	12V relay
5	Water flow sensor
6	Adaptor (12V,1 ampere)
7	Arduino Uno
8	Alphanumeric LCD (16x2)

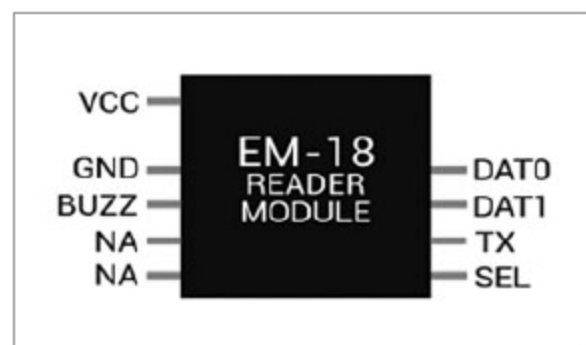


Fig. 3: EM-18 RFID reader



Fig. 4: Solenoid valve

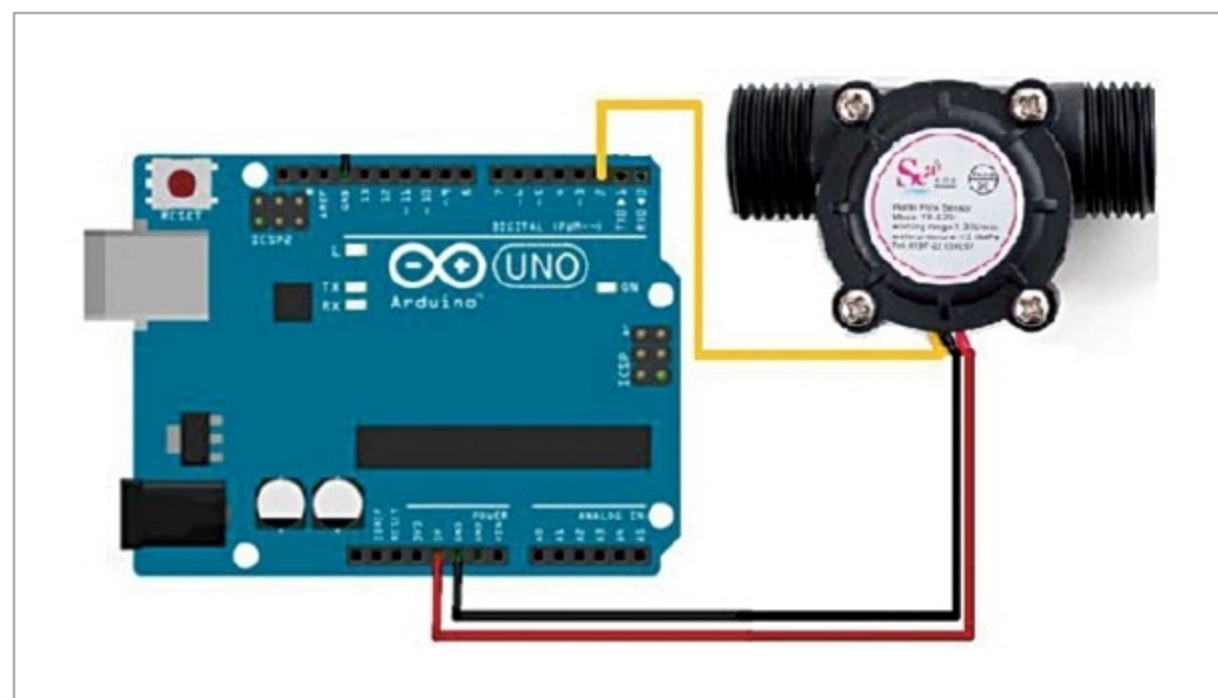


Fig. 5: Analogue flow sensor connected to Arduino

In this project, Arduino is the brain which controls the whole process. Analogue flow sensor connected to the Arduino board is shown in Fig. 5.

When liquid flows through the flow sensor, the liquid pushes against fins of the rotor, causing it to rotate. The shaft of the rotor is connected to a Hall effect sensor. It is an arrangement of a current flowing coil and a magnet connected to the shaft of the rotor, thus a voltage/pulse is induced as this rotor rotates. In this flow meter, for every litre of liquid passing through the sensor per minute, it outputs about 4.5 pulses.

Output pin of this flow sensor is connected to digital pin 2 of Arduino. When liquid flows, the output pulses are counted by the microcontroller in Arduino.

The total number of received pulses is converted into a specific unit like millilitre per second or litre per minute. This is due to the changing magnetic field caused by the magnet attached to the rotor shaft. Here, the flow rate in litres per minute (L/min.) is calculated using a simple conversion formula.

Pin Tx of EM-18 reader is connected to receive pin Rx of Arduino. When RFID tag swipes across EM-18 reader, it sends data to Arduino. If it matches with the programmed data, it sends a signal to the input of the relay driver consisting of transistor BC547 (T1). This input signal makes transistor to conduct, relay RL1 to get energised, which in turn connects 12V supply to solenoid via NO pin of relay and the solenoid gets energised. At the same time, "Place your pot/jug/cup" message is displayed on the first line of LCD1.

Energisation of solenoid allows the liquid to flow through the flow sensor. As the liquid starts to flow through the flow sensor, the amount of liquid being dispensed is displayed on the second line of LCD1.

When a predefined amount (programmed in the code) of liquid has passed through the flow sensor, the relay automatically gets de-energised and the flow of liquid stops. At the same time, "Remove your pot/jug/cup" message is displayed on the first line of LCD1.